

Peter Asanov

SENIOR SOFTWARE ENGINEER

Almere, Netherlands

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About

Senior software engineer with 20+ years of experience, including over a decade building real-time, data-intensive, and mission-critical distributed systems. Builds and owns cloud, on-premises, and edge platforms from architecture through production.

Skills

Distributed Systems HA/fault tolerance, scalability, partitioning/replication, EDA and messaging, consensus (Raft, Paxos), CAP trade-offs/consistency models

Development

C++11-23 Boost/Asio, Cap'n Proto, FlatBuffers, Seastar

Rust Tokio, hyper, PyO3/maturin

Go Gin, zap/slog

Python FastAPI/Pydantic, SQLAlchemy, Celery, NumPy, pandas, Polars

JVM Java/Scala/Kotlin, Spring Boot/Hibernate, Akka

JS/TS React, Vue, Vite/Vitest, tRPC, Prisma

Build Nix, CMake/Ninja, Conan, Maven, Bazel, Meson

Computer Systems

CPU/concurrency x86-64/ARM64 assembly, SIMD (SSE/NEON), memory models, memory and instruction ordering, fences/barriers, multithreading/multiprocessing
HW-aware optimization

Linux task scheduling, VM/paging, seccomp/SELinux, io_uring

Networking TCP/IP, UDP, DNS, routing/NAT, QoS, L4/L7 load balancing, virtual/overlay networks (Tailscale), eBPF/XDP/AF_XDP, Unix/raw sockets

Platform & Infrastructure

Cloud AWS (EC2/ECS/Lambda), Azure (VM/Functions), GCP (GCE/App Engine/Functions)

Virtualization/containers Proxmox, QEMU/KVM, Cloud Hypervisor, LXC/Incus, Docker Engine/containerd

Bare-metal/edge iPXE, Matchbox (Terraform provider), Flatcar/systemd-sysext, Keylime (TPM 2.0 attestation)

Kubernetes Talos Linux, k3s, EKS, AKS, GKE, OpenShift, Helm, Kustomize
Gateway API, Cilium (CNI), kube-vip, MetalLB
Strimzi, CloudNativePG, cert-manager, Trivy Operator, Prometheus Operator, kube-state-metrics

IaC Terraform, Tofu Controller, Ansible, CDK, CloudFormation, Bicep/ARM templates, NixOps4

CI/CD & GitOps Flux CD, Argo CD, GitHub Actions, GitLab CI/CD, Azure DevOps

Observability Prometheus, Grafana/LGTM, OTel/OTLP, CloudWatch/X-Ray, Azure Monitor/App Insights

IAM/IdPs OAuth 2.0/OIDC (PKCE/mTLS/DPoP), SAML, Entra ID/Azure AD, Cognito, Auth0

Security design trust models and Zero Trust, PoLP, authz models (ABAC, RBAC), isolation, audit

Data

DBs Postgres (PostGIS, TimescaleDB), DynamoDB, MySQL, Cassandra, MongoDB, DuckDB

KV/cache Redis, etcd, RocksDB, Valkey

Object storage S3, MinIO, RustFS, Azure Blob, GCS

Messaging Kafka, RabbitMQ, Pulsar

Pipelines Airflow, Spark/PySpark, Flink, NiFi

AI/ML

NLP

POS tagging, lemmatization, NER, NLTK, spaCy, Stanza

LLM integration

RAG, retrieval-ready data, agent orchestration, stateful multi-turn workflows

Cryptography

Curve25519, ECDH, EdDSA, Ristretto, ZK-SNARKs/ZK-STARKs

FOSS contributions



Tailscale

 tailscale/tailscale

Zero Trust identity-based connectivity platform

- Unix-domain socket support for Tailscale Serve, exposing local HTTP services over the tailnet without a TCP bridge.



ipsw

 blacktop/ipsw

iOS/macOS Research Swiss Army Knife

- dyld shared-cache fixes: address mapping, v4 rebasing and v5 pointer decoding.
- Objective-C scanning and ExclaveKit image resolution.
- Internal-symbol demangling for Mach-O disassembly.

Work Experience

Senior Software Engineer | Platform Architecture & Security

Netherlands

NEBUL

Started in Aug 2026

Own the architecture, planning and implementation of a declarative systems management platform for Nebul's datacenter infrastructure, designed for isolated networks and combining configuration enforcement, endpoint security and compliance reporting.

Contributions:

- Consolidate over 30 ADRs into a coherent system specification by building NLP-assisted pipeline for architectural review and decision traceability.
- Unify and extend existing security controls into a system-wide model, defining trust relationships and enforcing least privilege across service and operating-system boundaries. Validate process-supervision guarantees through crash-injection tests. Design integration with Nebul's node provisioning platform to anchor Arc enrollment in each node's TPM 2.0 root of trust.
- Design operational logging and audit flows, including durable central collection of node audit records and verified server acknowledgement of operation intent before sensitive changes.
- Establish the engineering foundation: reproducible builds, CI and dev workflow.
- Prototype secure agent communication over outbound connections, supporting isolated networks without inbound management access.

Tech: Go, Protobuf Editions/Buf/gRPC; Nix/flake-parts, GitLab CI, Talos Linux/Argo CD; mTLS, QUIC/WebTransport, HTTP/2 CONNECT fallback; OpenBao (PKI/Transit), Ed25519 signing, cosign, TPM 2.0 enrollment design; bubblewrap (namespaces/seccomp), cgroup v2, SELinux (confinement research); Python/spaCy, NetworkX, Markdown AST

Senior Software Engineer

Rotterdam, Netherlands

THE OCEAN CLEANUP

Mar 2026 – Jul 2026

Owned, re-architected, and shipped the infrastructure for Onboard Compute, an edge platform for UAV and vessel data ingestion, visualization, operational monitoring, and offshore cleanup planning.

- Scaled k3s from one node to a three-server HA cluster with an embedded etcd quorum; added Kubernetes API endpoint failover and an ingress VIP reachable from both the onboard LAN and tailnet.
- Introduced GitOps for in-cluster state and Ansible for node provisioning, access management, and k3s/Flux bootstrap. Secured runtime secrets with SOPS; validated fully rendered Flux trees with Kubelinter and strict kubeconform checks against pinned schemas; and built a topology-equivalent Tart/k3s environment for validation without field-cluster access.
- Authored an internal RFC to standardize Gateway API routing across edge and cloud systems and developed a Talos Linux PoC for immutable, API-managed edge nodes.

- Improved map legibility with higher-contrast density bands and clearer hierarchy between vessel and route layers.
- Instrumented UAV ingestion for Zenoh freshness and forwarding, Redis Streams/DLQ processing, and end-to-end latency; added alerts for Flux reconciliation, ingress health, and remote-write failures.
- Enabled unmodified amd64 workloads on Apple Silicon by replacing k3s's embedded containerd with Docker Engine and registering Rosetta through binfmt_misc, bypassing an OCI runtime/seccomp incompatibility.

Platform: k3s, Flux CD, kube-vip, MetalLB (L2), Tailscale, Traefik, CloudNativePG, PostgreSQL (PostGIS, TimescaleDB), Redis Streams, Zenoh, Prometheus/Grafana, RustFS

Tooling & app: Ansible, Kustomize, SOPS, KubeLint, kubeconform, Docker Engine, Tart/Rosetta, Nix/flake-parts, Python, TypeScript/Vue

Senior Software Engineer | Partner Integrations

Amsterdam, Netherlands

Vox AI

Jul 2025 – Oct 2025

Shipped backend integrations for a voice-AI drive-thru platform, working directly with founders.

- Delivered an OAuth 2.0/OIDC integration bridging AWS Cognito and a third-party DSP protocol within two weeks, enforcing PKCE while handling non-compliant partner clients.
- Built a multi-tenant location-management API on DynamoDB; added correlation IDs, X-Ray, Lambda Powertools, and EventBridge/SQS monitoring across authorization, token, and webhook flows.
- Structured the CDK stack to avoid 15–40 minutes of Cognito/CloudFront reprovisioning per iteration.
- Built a deterministic POS-to-RAG transformation pipeline that resolved ambiguity, circular dependencies, and orphaned references while remaining stable across upstream schema changes; integrated data-quality checks into CI.

Platform: AWS Lambda, API Gateway, Cognito, DynamoDB, X-Ray, CloudWatch, EventBridge, SQS, CDK

App stack: TypeScript, Python 3.12+, Pydantic, FastAPI, Mangum, Moto, pytest/pytest-asyncio, Vitest, HTTPX, uv

Senior Software Engineer

Netherlands

SELECTED CONTRACTS

2022 – Present

- **Deliverect:** extended the data model and API behavior of a restaurant-automation PaaS.
Stack: Python, Eve, Celery, GKE, TypeScript/React, Redis, MinIO, Terraform, Helm
 - **Ahold Delhaize (Albert Heijn):** built an online data ingestion platform for internal compliance and R&D analytics.
Stack: Go, AKS, Strimzi/Kafka, Argo CD, Airflow, Spark, Terraform, Helm
 - **Private client:** built taxi-terminal telemetry ingestion for compliance and fraud analysis.
Stack: Go, AKS, Kafka, PostgreSQL, Terraform, Helm
 - **LEGO:** built internal IAM and identity-provider functionality.
Stack: Azure Functions, SAML IdP, Python
 - **Private client:** built a cloud video-conversion service.
Stack: C++20, CMake/Ninja, Go, gRPC-Gateway, FFmpeg, EKS, CloudWatch, Airflow, Nix, Terranix, Helm
 - **Private client:** built a custom video-calling service.
Stack: C++23, CMake/Ninja, Asio, libdatachannel (WebRTC), Nix, Ansible
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Core Software Engineer

Haarlem, Netherlands

AIMMS B.V.

Nov 2019 – Sep 2021

Built core platform and migration tooling as a member of the AIMMS PRO core team:

- Migrated Java and Scala backend services from ECS to AKS; configured landing-zone RBAC, bootstrapped Terraform state, and delivered supporting IaC.
- Built a C++17 microservice template with an API-first protobuf DSL for gRPC and REST surfaces; contributed to Conan/Artifactory developer-platform infrastructure; built PoCs and delivered demos for the tooling.
- Improved cost efficiency and throughput predictability for compute-heavy, nondeterministic optimization workloads by implementing Kubernetes-native bin packing with resource requests/limits, affinity, priority classes, pre-warmed capacity, Cluster Autoscaler, and taints/tolerations.

- Built migration tooling for sensitive, business-critical customer data in Amazon RDS and a declarative-schema CRUD generator for administrative and internal tools.

Stack: ECS, EC2, RDS, CloudWatch, Go, Python, Java, Scala, C++17, TypeScript, React, Conan, Artifactory, CMake, Ninja, gRPC-Gateway, Kubernetes (AKS), Application Insights, Terraform, Helm

Senior Software Engineer

Amsterdam, Netherlands

HVR SOFTWARE (ACQUIRED BY FIVETRAN)

Jan 2018 – Oct 2019

- Extended HVR's high-throughput, log-based replication engine; reverse-engineered Oracle histogram metadata for custom sharding and built a Go REST service for cloud integration.
- Rewrote a core feature-flag transpiler used in HVR's CI/CD pipeline, reducing runtime from over one minute to five seconds on a critical build path.

Stack: C99, Python, Go, RDBMS and big-data integrations, Oracle (histogram internals), Airflow, Spark

Software Engineer

Eindhoven, Netherlands

MAPSCAPE B.V.

2017

Implemented backend components of the Mapscape NDS compiler for raw map data parsing and introduced parallel parsing to reduce processing time.

Stack: C++11/14, Boost, GTest, SQLite, shell, Python, Airflow

Senior Software Engineer

St. Petersburg, Russia

NEW CLOUD TECHNOLOGIES

2014 – 2017

Built collaboration-engine features for an on-premises office suite and developed its spreadsheet client for Samsung Tizen.

- Independently built a cross-platform FreeType-based font-rendering engine with subpixel positioning for scalable document layouts and integrated it into a large enterprise codebase.
- Developed advanced spreadsheet editing for Samsung Tizen using a declarative, compile-time state machine to avoid dynamic dispatch overhead and resolve editing edge cases observed in competing web suites.

Stack: C++14, CMake, Emscripten, Objective-C, Qt, SWIG, Kubernetes

Senior Software Engineer

St. Petersburg, Russia

AIS

2014

Built C++11 microservices for the banking and exchange subsystem of a cryptocurrency mining pool:

- Designed an Asio-based service framework and implemented all C++11 banking services, including multi-stage, fault-tolerant transactions.
- Protected financial state with append-only storage, two-factor transactions, circuit breakers, alerting, rollback paths, and extensive testing.
- Shipped the zero-to-one production release in under two months with a three-engineer team, owning the C++ subsystem end to end.

Stack: C++11, Core Java, Groovy, Ruby (Chef)

Education

B.Sc. in Computer Science and Engineering

Novosibirsk, Russia

NOVOSIBIRSK STATE TECHNICAL UNIVERSITY (NSTU)

2002 – 2007

Previous work history

AVAILABLE UPON REQUEST

2005 – 2014